Paul Fessler
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DTE

10 CFR 50.73

July 26, 2019
NRC-19-0053

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2019-003

Pursuant to 10 CFR 50.73(a)(2)(v)(C), DTE Electric Company (DTE) is submitting LER No. 2019-003.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Jason R. Haas, Manager – Nuclear Licensing, at (734) 586-5076.

Sincerely,

Paul Fessler
Senior Vice President and CNO

Enclosure: Licensee Event Report No. 2019-003, Secondary Containment Declared Inoperable Due to Opening Both Airlock Doors

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov

**Enclosure to**
**NRC-19-0053**

**Fermi 2 NRC Docket No. 50-341**
**Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2019-003**
**Secondary Containment Declared Inoperable Due to Opening Both Airlock Doors**

U.S. NUCLEAR REGULATORY COMMISSION

# LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)



| 1. Facility Name | 2. Docket Number | 3. Page |
|---|---|---|
| Fermi 2 | 05000 341 | 1 OF 3 |

**4. Title**

Secondary Containment Declared Inoperable Due to Opening Both Airlock Doors

| 5. Event Date | | | 6. LER Number | | | 7. Report Date | | | 8. Other Facilities Involved | |
|---|---|---|---|---|---|---|---|---|---|---|
| Month | Day | Year | Year | Sequential Number | Rev No. | Month | Day | Year | Facility Name N/A | Docket Number 05000 |
| 05 | 29 | 2019 | 2019 | 003 | 00 | 07 | 26 | 2019 | Facility Name N/A | Docket Number 05000 |

**9. Operating Mode:** 1

**10. Power Level:** 100

**11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)***

| | | | |
|---|---|---|---|
| ☐ 20.2201(b) | ☐ 20.2203(a)(3)(i) | ☐ 50.73(a)(2)(ii)(A) | ☐ 50.73(a)(2)(viii)(A) |
| ☐ 20.2201(d) | ☐ 20.2203(a)(3)(ii) | ☐ 50.73(a)(2)(ii)(B) | ☐ 50.73(a)(2)(viii)(B) |
| ☐ 20.2203(a)(1) | ☐ 20.2203(a)(4) | ☐ 50.73(a)(2)(iii) | ☐ 50.73(a)(2)(ix)(A) |
| ☐ 20.2203(a)(2)(i) | ☐ 50.36(c)(1)(i)(A) | ☐ 50.73(a)(2)(iv)(A) | ☐ 50.73(a)(2)(x) |
| ☐ 20.2203(a)(2)(ii) | ☐ 50.36(c)(1)(ii)(A) | ☐ 50.73(a)(2)(v)(A) | ☐ 73.71(a)(4) |
| ☐ 20.2203(a)(2)(iii) | ☐ 50.36(c)(2) | ☐ 50.73(a)(2)(v)(B) | ☐ 73.71(a)(5) |
| ☐ 20.2203(a)(2)(iv) | ☐ 50.46(a)(3)(ii) | ☑ 50.73(a)(2)(v)(C) | ☐ 73.77(a)(1) |
| ☐ 20.2203(a)(2)(v) | ☐ 50.73(a)(2)(i)(A) | ☐ 50.73(a)(2)(v)(D) | ☐ 73.77(a)(2)(i) |
| ☐ 20.2203(a)(2)(vi) | ☐ 50.73(a)(2)(i)(B) | ☐ 50.73(a)(2)(vii) | ☐ 73.77(a)(2)(ii) |
| | ☐ 50.73(a)(2)(i)(C) | ☐ Other (Specify in Abstract below or in NRC Form 366A) | |

**12. Licensee Contact for this LER**

| Licensee Contact | Telephone Number (Include Area Code) |
|---|---|
| Fermi 2 / Jason Haas - Manager, Nuclear Licensing | (734) 586-5076 |

**13. Complete One Line for each Component Failure Described in this Report**

| Cause | System | Component | Manufacturer | Reportable to ICES | Cause | System | Component | Manufacturer | Reportable to ICES |
|---|---|---|---|---|---|---|---|---|---|
| A | NH | DR | N/A | Y | N/A | N/A | N/A | N/A | N/A |

| 14. Supplemental Report Expected | | 15. Expected Submission Date | Month | Day | Year |
|---|---|---|---|---|---|
| ☐ Yes (If yes, complete 15. Expected Submission Date) | ☐ No | | | | |

**Abstract** (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On May 29, 2019 at 2210 hours, during Security Force on Force Practice Drills both doors in the Secondary Containment (SC) air lock on the reactor building first floor were opened simultaneously for a period of approximately two seconds. During the drill exercise, the Controller failed to maintain the access control measures at the outer air lock door which resulted in additional drill personnel accessing through the door.

The condition resulted in Technical Specification (TS) Surveillance Requirement (SR) 3.6.4.1.3 not being met. The doors were secured immediately. SC pressure during that time remained unchanged and within TS limits. There were no radiological releases associated with this event. Declaring Secondary Containment inoperable for not meeting TS SR 3.6.4.1.3 is reportable under 10 CFR 50.72(b)(3)(v)(C). This written report is required under 10 CFR 50.73(a)(2)(v)(C) as an event or condition that could have prevented the fulfillment of a safety function needed to control the release of radioactive material.

| 1. FACILITY NAME | 2. DOCKET NUMBER | 3. LER NUMBER YEAR | SEQUENTIAL NUMBER | REV NO. |
|---|---|---|---|---|
| Fermi 2 | 05000- 341 | 2019 | - 003 | - 00 |

NARRATIVE

INITIAL PLANT CONDITIONS
Mode 1
Reactor Power 100%

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On May 29, 2019 at 2210 hours, during Security Force on Force (FOF) Practice Drills, Secondary Containment (SC) [[NH]] doors [[DR]] at the Reactor Building (RBD08) first floor were opened at the same time.

As part of the planned sequence of events, the “Drill Adversaries” entered and took position at the inner air lock door and propped the door open. The drill Controller stayed back behind the outer air lock door to stop personnel from accessing the door. During the drill exercise, the Controller failed to maintain the access control measures at the outer air lock door which resulted in additional drill personnel accessing through the door creating a loss of SC. On recognition of the condition, the inner air lock door was immediately secured.

The Main Control Room was notified that both doors in the SC air lock on the Reactor Building first floor (RBD08) were opened simultaneously for a period of approximately two seconds. This resulted in Technical Specification (TS) Surveillance Requirement (SR) 3.6.4.1.3 not being met. SC pressure remained unchanged and within TS limits.

There were no radiological releases associated with this event. Declaring SC inoperable for not meeting TS SR 3.6.4.1.3 is reportable under 10 CFR 50.72(b)(3)(v)(C). This written report is required under 10CFR50.73(a)(2)(v)(C) as an event or condition that could have prevented the fulfillment of a safety function needed to control the release of radioactive material. The doors were closed in approximately 2 seconds and SC was restored.

Event notification 54094 was submitted at 0010 hours on May 30, 2019.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety. The SC air lock doors (RBD08) were opened for approximately two seconds. When both air lock doors were open for approximately 2 seconds on May 29, negative pressure was maintained in SC. The SC vacuum remained within the TS operability limit of greater than or equal to 0.125 inches of vacuum water gauge per TS Surveillance Requirement (SR) 3.6.4.1.1. The Standby Gas Treatment System did not start. There was no unmonitored release during the event.

| 1. FACILITY NAME | 2. DOCKET NUMBER | 3. LER NUMBER: YEAR | SEQUENTIAL NUMBER | REV NO. |
|---|---|---|---|---|
| Fermi 2 | 05000- 341 | 2019 | - 003 | - 00 |

NARRATIVE

CAUSE OF THE EVENT

As part of the planned sequence of events, the "Drill Adversaries" carded through RBD08 to take up a position at the inner air-lock door (propping the door open); while the drill controller stayed back behind the outer air-lock door to stop personnel from accessing the door. During the course of the drill play, the controller failed to maintain the access control measures at the outer air lock door which resulted in additional drill personnel accessing through the door creating a loss of SC for approximately two seconds.

The direct cause evaluation concluded that the planned access control measures failed when the drill controller (utility – unlicensed) failed to maintain situational awareness and was involved in a conversation detailing the simulated drill configuration of the door. During the conversation an additional drill player (utility – unlicensed) accessed the door resulting in the SC breach. The lack of situational awareness was a human performance error.

CORRECTIVE ACTIONS

The inner air lock door was immediately secured. The door was blocked for general personnel to access by the Shift Manager.

A human performance departmental event free day reset evaluation was completed. The event briefing sheet was presented and discussed within the security department. Security Management communicated to the staff the importance of maintaining situational awareness during critical tasks.

Security Procedure “Guidance for the Preparation of Force on Force” was revised to add additional guidance for the Drill Controllers to mitigate distractions during drills and exercises.

PREVIOUS OCCURRENCES

There was one previous occurrence that involved the human performance error of simultaneous opening of both air lock doors. The event was reported by Licensee Event Report (LER) No. 2019-002, Secondary Containment Airlock Doors Opened at the Same Time Resulting in a Condition Prohibited by Technical Specifications and involved a separate SC air lock.

The event described in this LER is different than LER 2019-02 because in that case the SC air lock doors had engineering controls that failed. The doors had an interlock that should have prevented opening both doors.

When both airlock doors were opened simultaneously the doors became stuck open for about 5 minutes until the latching pin was adjusted.

Both LERs described an element of a human performance shortfall that allowed both doors to be opened at the same time.